

Drug Search and Physician Hazard:
An Investigation into Addict Behavior and Policy Remedies

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Abstract

Keywords:

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1 Introduction

1.1 Background and Motivation

Over the last few decades, prescription drug abuse has become a significant and growing public health concern across the globe. The misuse of prescription drugs, specifically opioids, benzodiazepines, and stimulants, has led to the question of how to mitigate the practice of physician search. Physician search refers to the practice of patients seeking multiple doctors in the hope of "scoring" a prescription to continue their addiction. While there are preventative measures in place, like registries of offenders, the practice persists. Coinciding with this question, it would be of great use to ascertain the incentive for physicians to enable the misuse of drugs, gaining a repeat source of revenue.

This paper seeks to understand the interplay of addicts and physician search, in hope of policy remedies which are more effective than a list. Further, the analysis of prescriber responsibility is also of great importance. Being able to understand how morals and fiscal incentives contrast in this case could help to implement policy which negates the opportunity for prescribers to mis-prescribe a sensitive drug. Finally, this paper will look to gain an insight into potential rehabilitation remedies which may help take the onus off both addicts and physicians.

1.2 Problem Statement

To ascertain the interplay between physician and addict behavior, it would be of interest to model physician WTP, in a sense, with their reputation. That is, how much risk are physicians willing to onboard for the profits associated with increased and repeat prescriptions? This would be interesting for many reasons, since physicians have many reasons to uphold their standing from liability, licensure, and even word of mouth advertisement. The last one may be increased as addicts see a pathway to drug access, but that would also increase liability and risk of loss of their licensure.

For addicts, it is interesting to inquire about their search cost allocation, as well as habit based expenditure on health. Tolerance necessarily grows with prolonged use of drugs, thus modeling this sort of depletion of both search and health budgets is of interest to understand the impact on patients of finding a willing prescriber.

From these two avenues, I think there is a worthwhile question to answer. That is, what is the risk threshold physicians will take on in exchange for profit as well as understanding time and health costs applied to addicts once they find a willing prescriber.

1.3 Contributions

1.4 Paper Organization

2 Related Work

2.1 Previous Approaches

2.2 Limitations of Existing Work

3 Model

3.1 Addicts

There is a continuum of addicts in this model such that $\mathcal{A} = \{a_1, a_2, \dots, a_N\}$, each with an individual type $\gamma_i \in \{\gamma_l, \gamma_h\}$, corresponding to high or low predisposition to addiction. Each addict maximizes their utility over an infinite horizon, with discount factor β . Each addict also has a tolerance for risk of being cut off from search, r_i , which evolves according to their type and consumption behavior. Each period, they choose to search for a physician, and will successfully receive a prescription with probability $m(\theta)$, where θ is the measure of physicians willing to prescribe. Searching comes at a cost s_i , which depletes their health budget, and increases their risk of being cut off. Each period, their risk evolves according to $r'_i = r_i + \epsilon_a$, where $\epsilon_a \sim N(\mu_a, \sigma_a^2)$. Addicts optimize over the following: In the first period:

$$V_a^0(\gamma_i, r_i) = \{u(c_i) + \beta \mathbb{E}_t(1 - a(\theta))V_a(\gamma_i, r_i) + \beta \mathbb{E}_t a(\theta)V_a(\gamma_i, r'_i)\} \quad (1)$$

s.t.

$$u(c) = \frac{c_i^{1-\sigma}}{1-\sigma}$$

$$r'_i = r_i + \epsilon_a$$

In proceeding periods:

$$V_a^m(\gamma_i, r_i) = \max\{V_a^s(\gamma_i, r_i), V_a^f(\gamma_i, r'_i)\} \quad (2)$$

s.t.

$$V_a^s(\gamma_i, r_i) = \{u(c_i) + \beta V_a^m(\gamma_i, r'_i)\}$$

$$V_a^f(\gamma_i, r'_i) = \{u(c_i) + \mathbb{E}_t[\beta(1 - a(\theta))V_a(\gamma_i, r'_i) + \beta \mathbb{E}_t[a(\theta)V_a(\gamma_i, r'_i)]]\}$$

$$u(c) = \frac{c_i^{1-\sigma}}{1-\sigma}$$

$$r'_i = r_i + \epsilon_a$$

$$c_i = \gamma_i p_{it} - r_i^{1/\gamma_i} \times \sum s_{it}$$

Where V_a^0 is the value function for an addict in their first period of search, and V_a^m is the

value function for an addict in proceeding periods, if they choose to find a new doctor, V_a^f , or to stay with their current match, V_a^s . $u(c_i)$ is the utility from consumption, with CRRA parameter σ . c_i is comprised of prescription utility (weighted by tolerance), minus the addicts search costs augmented by risk. Addicts will always search if they are unmatched, and will choose to search again if their value from searching exceeds that of staying with their current physician, or they are cut off.

3.2 Doctors

There is a continuum of doctors such that $\mathcal{M} = \{m_1, m_2, \dots, m_K\}$ who value profit. They will exist in the model until they have overprescribed, taking on too much risk, $p_{jt} > r_{jt}$. Doctors can be matched with many addicts, with $p_{jt} = \sum_{i=1}^I c_{it}$, for all patients $I \subset \mathcal{A}$ matched to them. They will drop patients when their consumption needs threaten their risk constraint, thus allowing them to bring in a new, lower risk, patient. Doctors will prescribe to addicts with probability $m(\theta)$, where θ is the measure of physicians willing to prescribe. Each period, their risk evolves according to $r'_{jt} = r_{jt} + \epsilon_d$, where $\epsilon_d \sim N(\mu_d, \sigma_d^2)$.

The doctor's optimization is as follows

$$\begin{aligned} \max_{v_j} \quad & v_j - i_j r_{jt} \\ \text{s.t.} \quad & \\ v_j = & r_d^2 \sum_{i=1}^I p_{it} + \sum_{i=1}^I a_i \end{aligned} \tag{3}$$

Where v_j is a visit cost, which is initially a flat ξ and will evolve according to how many patients they prescribe to. i_j is a flat physician's insurance cost which is raised by amount of risk. This can be thought of as malpractice insurance, or something to that effect. Basically, they want to maximize visit revenue, net of their risk adjusted fixed cost.

Doctors who do not prescribe will have a different optimization, which is simply maximizing their revenue from visits, net of risk cost, which is 0 since they do not prescribe.

3.3 Estimation

This was estimated via VFI. Estimation procedure is as follows:

1. Initialize parameters for addicts and doctors, as well as initial value functions.
2. For each iteration:
 - (a) Solve the addict's dynamic programming problem using value function iteration.
 - (b) Solve the doctor's optimization problem given the current distribution of addicts.
 - (c) Update the measure of physicians willing to prescribe, θ , based on the current matches.
 - (d) Check for convergence of value functions and policies.

3. Once converged, simulate the model to generate synthetic data for analysis.
4. Simulate counterfactual scenarios to assess policy impacts.

4 Simulated Results

4.1 Baseline Simulation

In the baseline simulation, we set the following parameters:

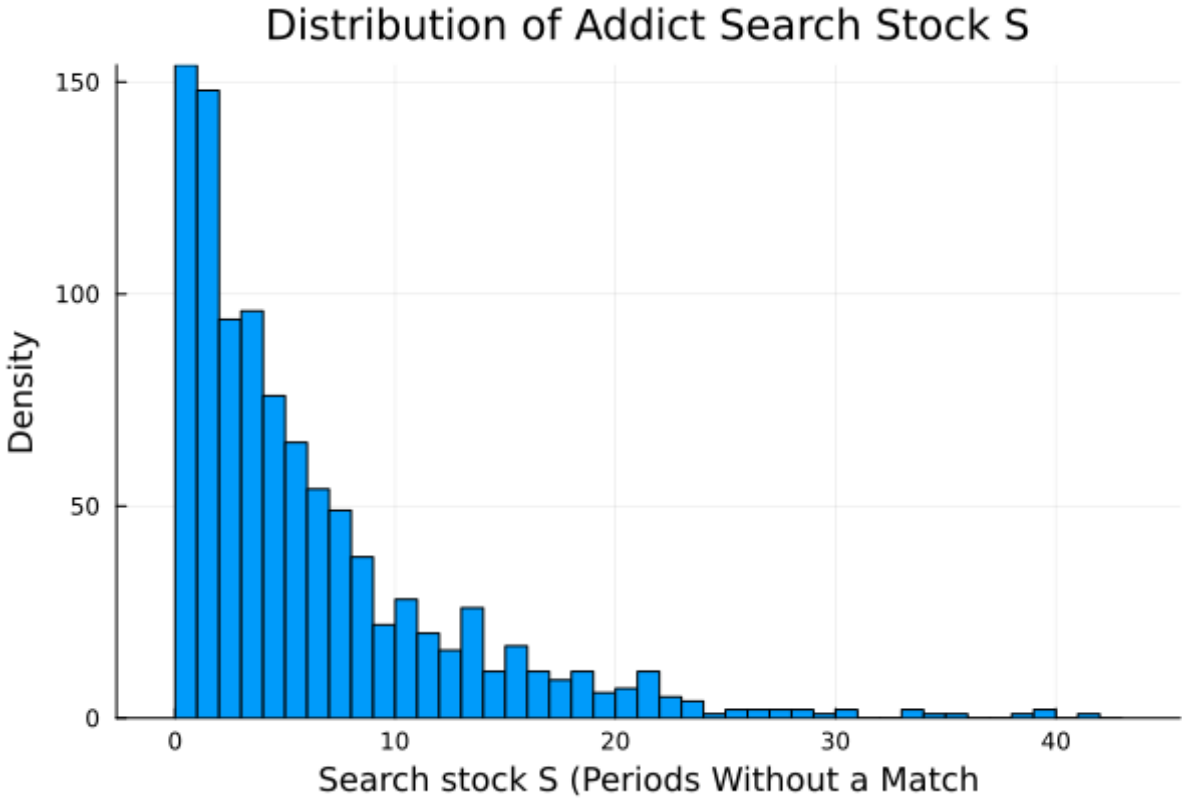
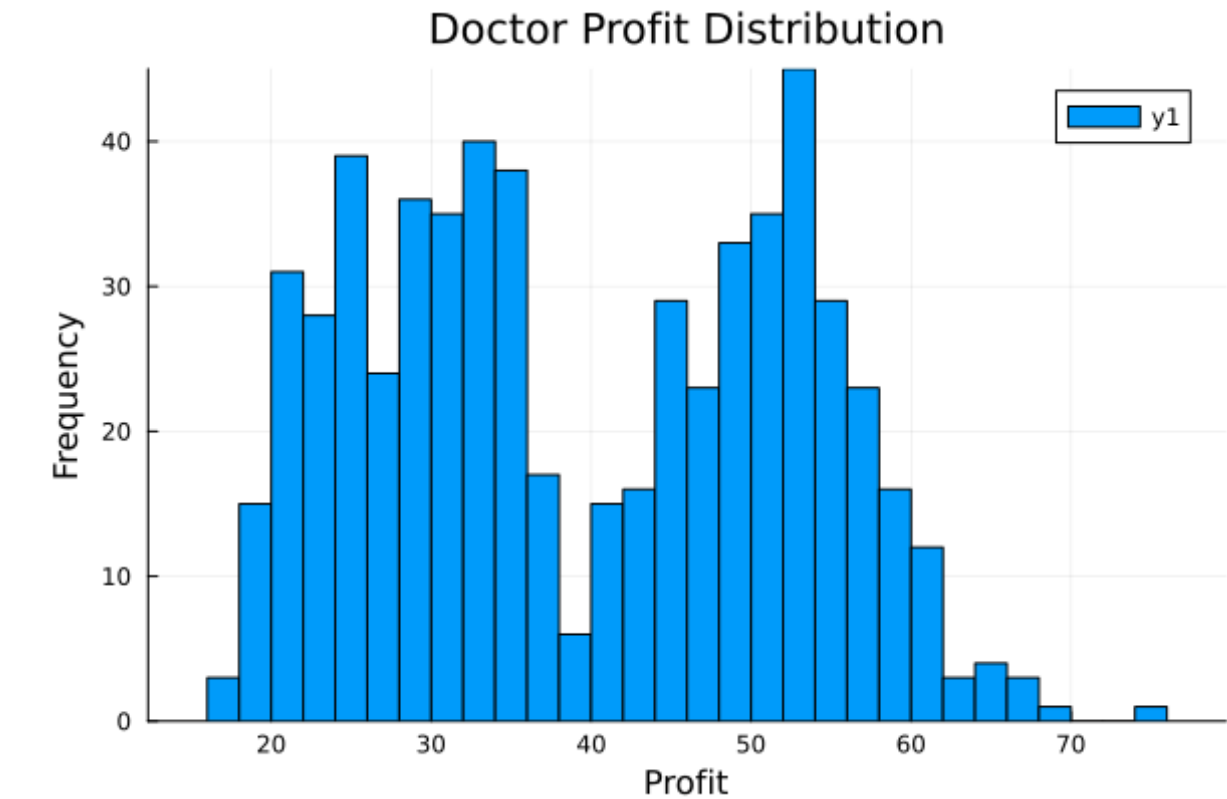
- Addict discount factor $\beta = 0.95$
- High addiction predisposition $\gamma_h = 1.5$ and low addiction predisposition $\gamma_l = 0.5$
- Probability of successful prescription $m(\theta) = 0.3$
- Search cost $s_i = 1$
- Risk caps: $r_d = \{10, 30\}$; $r_a = \{5, 15\}$
- Number of addicts $N = 1000$ and number of doctors $K = 600$

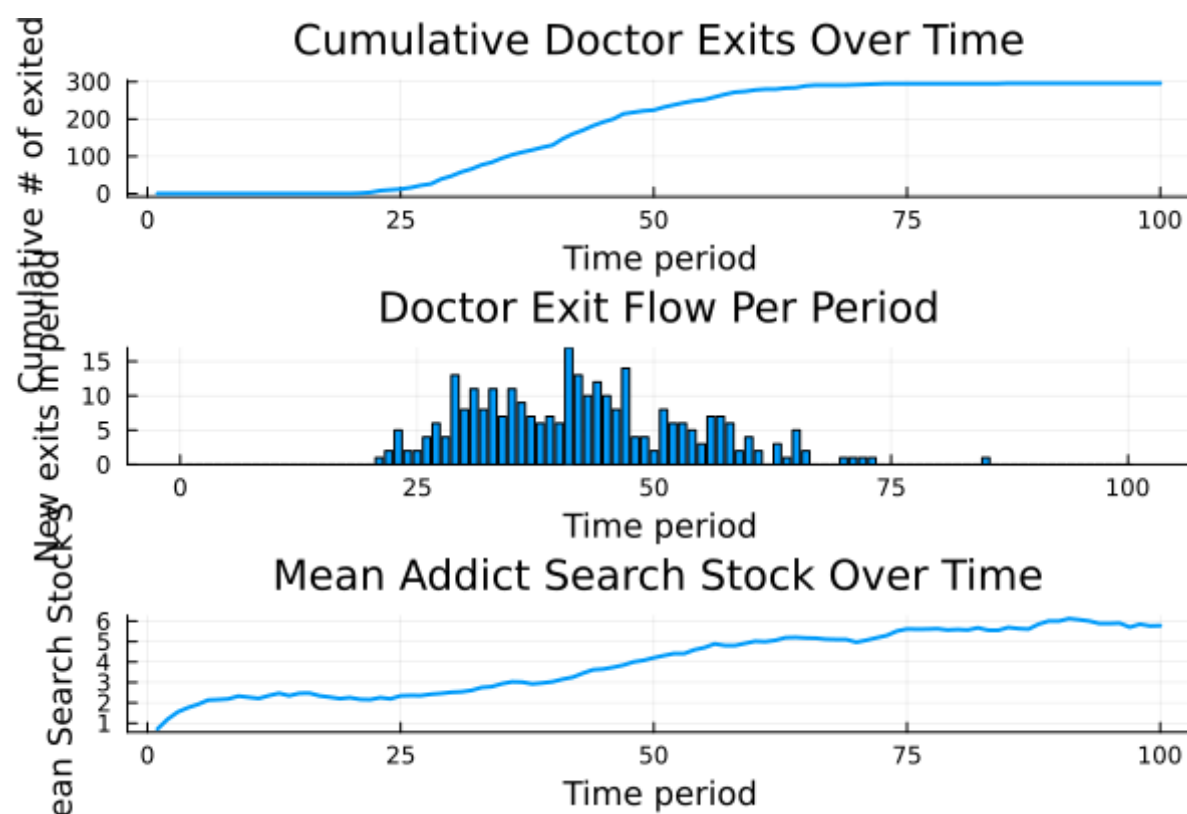
4.2 Policy Simulations

The main counterfactual changed risk for doctors, making the risk cap lower, simulating increased regulation and oversight. This was done to see how physicians would react to increased risk of prescribing, and how that would affect addict behavior.

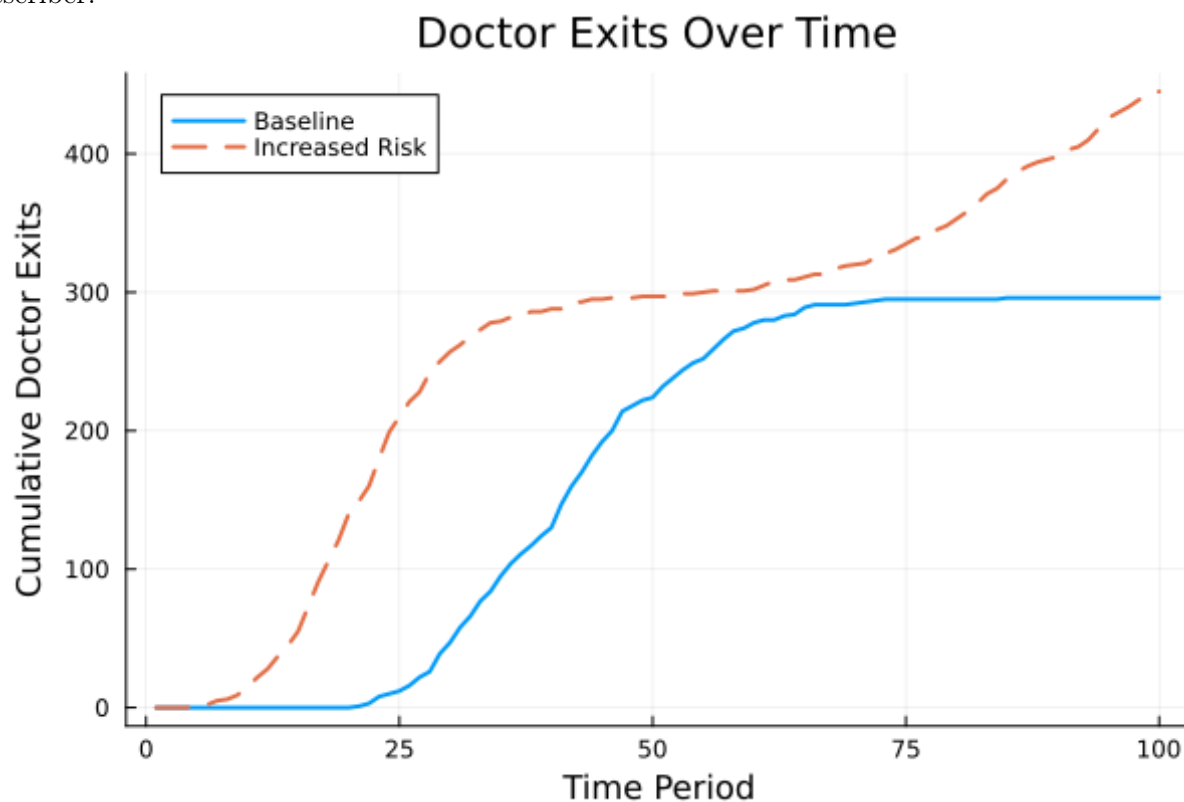
5 Discussion

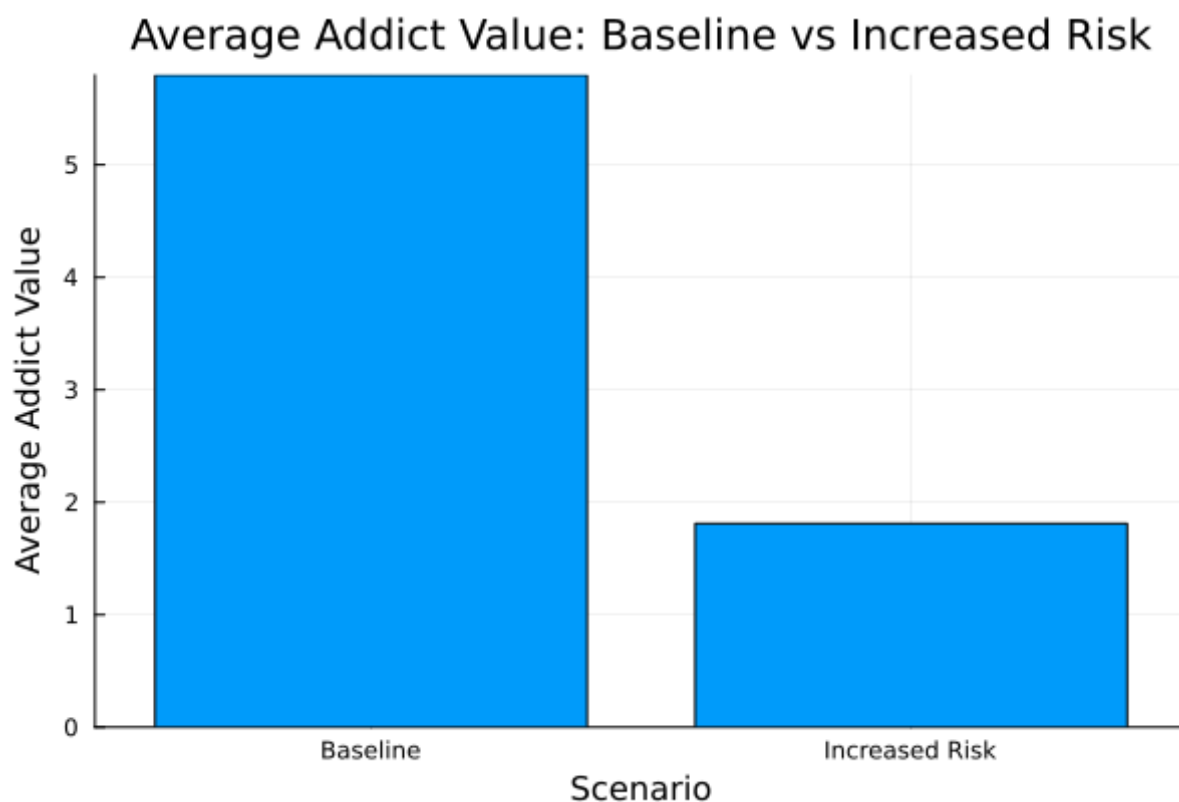
In the baseline, it can be seen that doctors are willing to take on increasing risk for about 25 periods, after which we begin to see exit from the prescribing market. This is likely due to the accumulation of risk from prescribing to addicts, who have increasing tolerance and thus require more prescriptions over time. Addicts, however, seem to have a relatively stable search pattern, with high initial search stock, which decreases after about 10 straight periods. Doctor profits take a bimodal distribution, with some doctors making very high profits, while others make very little. This is likely a side effect of only having two types of risk tolerance for doctors.



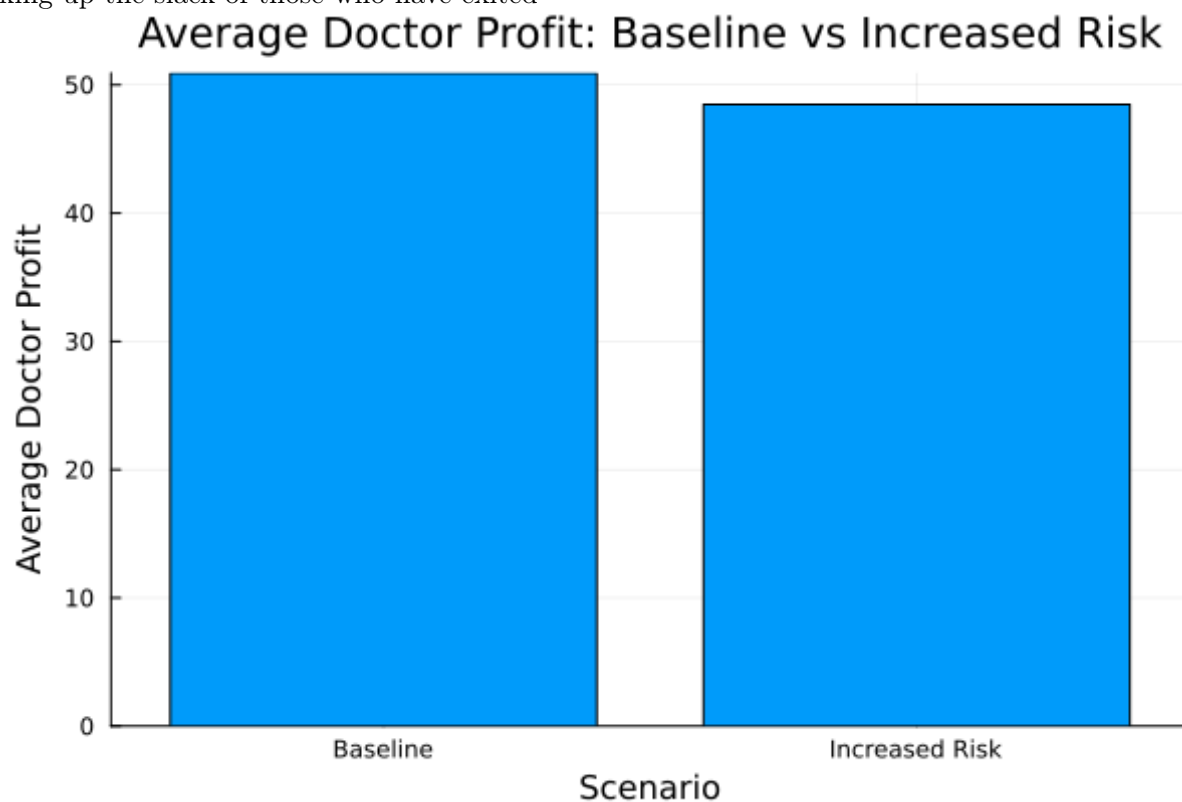


In the policy simulation, we see that doctors exit the prescribing market much earlier, and we see that addicts have significant lower value due to the increased difficulty in finding a willing prescriber.





Interestingly, we do not see a significant decrease in the profits of doctors in the counterfactual, likely due to the fact that those who remain prescribing are doing so at a higher rate, picking up the slack of those who have exited



6 Messy Data

When thinking about data that would be fruitful for this project, there are a few avenues that should be explored. First, and most obviously, claims data would allow me to get an idea of the search process of addicts. This would allow me to parse out how many physicians an addict visits before finding a willing prescriber, along with frequency of visits, and prescription amounts. These insights would allow me to recover γ and s_i for addicts, as well as give me a better idea of the distribution of addict types, instead of just high or low.

Next, survey data from Monitoring the Future dataset could be useful to understand the behavior of addicts, especially over time. This data would let me see how addicts' γ evolves over time as well, while also giving me a better sense of their budget constraints. Further, it may be helpful for understanding exit conditions of addicts, who have to leave due to cost or desire for more prescription consumption turning into illicit drug use.

Finally, physician data from state medical boards or the AMA could be useful to understand prescribing behavior. This data could help me understand how physicians' risk tolerance evolves over time, as well as give me a better idea of the distribution of physician types. Further, this data could help me understand exit conditions for physicians, who may leave the prescribing market due to risk of losing their license or malpractice suits.

These data sources together would provide a comprehensive view of the addict-physician interaction, allowing for more accurate parameter estimation and model validation. Additionally, I would be able to extend the model to include more sources of uncertainty and heterogeneity, leading to richer policy implications.